IANLP 2026 · AI EXCELLENCE LICENSE & MASTER IN ARTIFICIAL INTELLIGENCE

Teaching a computer to recognise *pollen*

A convolutional neural network that tells 21 plant species apart from a single microscope image of a pollen grain.

Youssef Charroud · Marwane Laamiri · Imad Najam

21 plant species it can name

655 microscope images studied

16 training rounds, with fine-tuning

92% correct on images it had never seen

01 Problem

Identifying pollen grains by eye is slow and demands rare expert palynologists. With 21 visually similar species, manual sorting does not scale. Can a machine learn to do it from a single microscope image?

02 Objectives

- Automate identification of 21 pollen species
- Reach expert-level accuracy with no human input
- Keep the model light — trainable in minutes

IN PLAIN WORDS

From a blurry grey speck, the model names the right flower more than **9 times out of 10** — a task most people could not do at all.

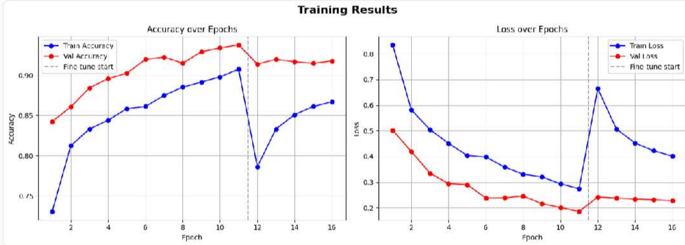
03 Methodology

Each image flows down a chain of steps. Every step throws away noise and keeps the clues that matter — until only the species is left.

- **THE PHOTO** — A 224×224 microscope picture
- **TIDY UP** — Brightness levels are evened out
- **FIND EDGES** — First conv. layer spots outlines & shapes
- **FIND TEXTURE** — Second layer reads surface patterns
- **WEIGH CLUES** — A dense network weighs the evidence
- **THE ANSWER** — The species, with a confidence %

04 Results

Across 16 training rounds — including a fine-tuning phase — accuracy climbs steadily and loss falls. The model reaches about 92% accuracy on new, unseen images.

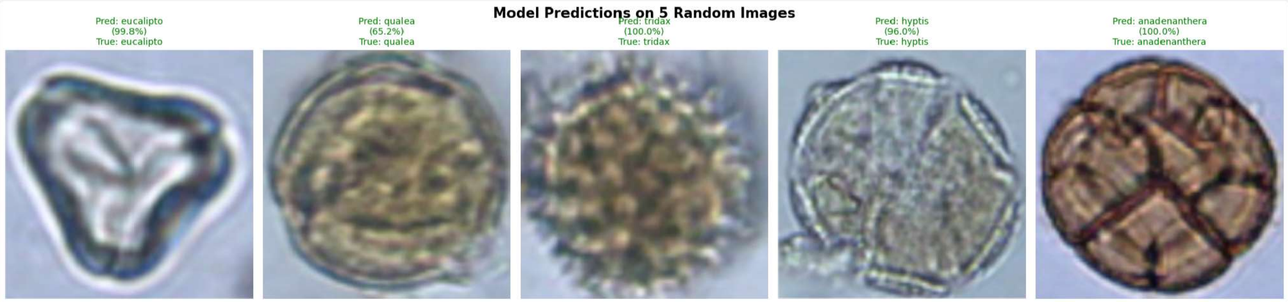


Accuracy and loss over the 16 training epochs — training vs. validation.

05 Impact

Faster allergy and pollen-season forecasting · honey-origin and food verification · biodiversity and crop monitoring — bringing expert pollen identification to any lab or field device.

06 Predictions on unseen grains



Tested on five random grains it had never seen — predicted species and confidence against the true label. All five correct.

The 21 species it tells apart

Anadenanthera · *Arecaceae* · *Arrabidaea* · *Cecropia* · *Chromolaena* · *Combretum* · *Croton* · *Dipteryx* · *Eucalipto* · *Faramea* · *Hyptis* · *Mabea* · *Matayba* · *Mimosa* · *Myrcia* · *Protium* · *Qualea* · *Schinus* · *Serjania* · *Syagrus* · *Tridax*